

# DIP-SMC-PSO: Technical Progress Memo

Model choices, algorithm design, and experimental results

February 2026 — Repository: [github.com/theSadeQ/dip-smc-psy](https://github.com/theSadeQ/dip-smc-psy) — Status: Research complete, thesis 90%

## Contents

|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| <b>1</b> | <b>System Model: Equations and Parameter Choices</b>                | <b>2</b>  |
| 1.1      | State Vector and Coordinate Definition . . . . .                    | 2         |
| 1.2      | Equations of Motion . . . . .                                       | 2         |
| 1.3      | Physical Parameters . . . . .                                       | 3         |
| 1.4      | Model Selection: Simplified vs. Full Nonlinear . . . . .            | 3         |
| <b>2</b> | <b>Sliding Mode Control: Surface Design and Control Laws</b>        | <b>4</b>  |
| 2.1      | Sliding Surface . . . . .                                           | 4         |
| 2.2      | Controller Variants: Control Laws . . . . .                         | 4         |
| 2.2.1    | Classical SMC: Saturation Function and Equivalent Control . . . . . | 4         |
| 2.2.2    | Super-Twisting Algorithm (STA-SMC) . . . . .                        | 5         |
| 2.2.3    | Adaptive SMC . . . . .                                              | 5         |
| 2.2.4    | Hybrid Adaptive STA-SMC . . . . .                                   | 6         |
| <b>3</b> | <b>PSO Algorithm: Configuration and Methodology</b>                 | <b>6</b>  |
| 3.1      | Algorithm Parameters . . . . .                                      | 6         |
| 3.2      | Fitness Function . . . . .                                          | 6         |
| 3.3      | Search Bounds per Controller . . . . .                              | 6         |
| 3.4      | Robust PSO: Scenario-Based Optimization . . . . .                   | 7         |
| 3.5      | PSO-Optimized Gains (MT-8 Robust Tuning) . . . . .                  | 7         |
| <b>4</b> | <b>Lyapunov Stability: Proofs and Validation</b>                    | <b>8</b>  |
| 4.1      | Classical SMC . . . . .                                             | 8         |
| 4.2      | Super-Twisting (STA-SMC) . . . . .                                  | 8         |
| 4.3      | Adaptive SMC . . . . .                                              | 9         |
| 4.4      | Hybrid Adaptive STA-SMC . . . . .                                   | 9         |
| 4.5      | Swing-Up SMC . . . . .                                              | 9         |
| 4.6      | MPC . . . . .                                                       | 9         |
| 4.7      | Summary Table . . . . .                                             | 10        |
| <b>5</b> | <b>Monte Carlo Benchmark Results</b>                                | <b>10</b> |
| 5.1      | Experimental Setup . . . . .                                        | 10        |
| 5.2      | Primary Performance Metrics . . . . .                               | 11        |
| 5.3      | Chattering Analysis (FFT) . . . . .                                 | 11        |
| 5.4      | Statistical Significance (Welch's $t$ -test) . . . . .              | 12        |
| 5.5      | Energy Trade-off Analysis . . . . .                                 | 12        |
| <b>6</b> | <b>Boundary Layer Study: MT-6 Honest Result</b>                     | <b>12</b> |
| <b>7</b> | <b>Robustness Analysis</b>                                          | <b>13</b> |
| 7.1      | Model Uncertainty (LT-6) . . . . .                                  | 13        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| 7.2      | Disturbance Rejection (MT-8) . . . . .                     | 13        |
| <b>8</b> | <b>Thesis and Paper Status</b>                             | <b>14</b> |
| 8.1      | Research Paper (LT-7): Submission-Ready . . . . .          | 14        |
| 8.2      | Thesis: 90/100 (Near Submission-Ready) . . . . .           | 14        |
| <b>A</b> | <b>Controller Gains Reference (PSO-Tuned, Operational)</b> | <b>14</b> |
| <b>B</b> | <b>Key Citations</b>                                       | <b>14</b> |

## 1 System Model: Equations and Parameter Choices

### 1.1 State Vector and Coordinate Definition

The DIP state is a 6-vector:

$$\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \in \mathbb{R}^6 \quad (1)$$

where  $x$  is cart position,  $\theta_1$  and  $\theta_2$  are pendulum link angles (from vertical), and dots denote time derivatives.

### 1.2 Equations of Motion

The full Lagrangian model is:

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_{\text{fric}} = \mathbf{B} u \quad (2)$$

where  $\mathbf{q} = [x, \theta_1, \theta_2]^\top$ ,  $u \in [-150, 150]$  N is the cart force, and  $\mathbf{B} = [1, 0, 0]^\top$  selects the actuated direction.

**Inertia matrix  $\mathbf{M}(\mathbf{q})$  ( $3 \times 3$ , symmetric, configuration-dependent):** The matrix varies 40–60% across the workspace. All 9 entries are (symmetric so  $M_{ij} = M_{ji}$ ):

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} M + m_1 + m_2 & (m_1 l_{c1} + m_2 l_1) \cos \theta_1 + m_2 l_{c2} \cos \theta_2 & m_2 l_{c2} \cos \theta_2 \\ \cdot & m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2) + I_1 + I_2 + 2m_2 l_1 l_{c2} \cos(\theta_1 - \theta_2) & m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} \cos(\theta_1 - \theta_2) \\ \cdot & \cdot & m_2 l_{c2}^2 + I_2 \end{bmatrix} \quad (3)$$

where  $\cdot$  denotes the symmetric entry.  $M_{22}$  contains the  $2m_2 l_1 l_{c2} \cos(\theta_1 - \theta_2)$  coupling term that makes the matrix strongly configuration-dependent near the horizontal positions.

**Coriolis/centrifugal matrix  $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$  (non-zero entries):** Computed from the Christoffel symbols of (3); all zero entries omitted:

$$\begin{aligned} C_{01} &= -(m_1 l_{c1} + m_2 l_1) \sin \theta_1 \dot{\theta}_1 - m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{02} &= -m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{11} &= C_{12} = -m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_2 \\ C_{21} &= +m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_1 \end{aligned} \quad (4)$$

(indices 0,1,2 correspond to  $x, \theta_1, \theta_2$ ). A small gyroscopic coupling term  $g_c(\dot{\theta}_1 + \dot{\theta}_2)$  is added to  $C_{01}$  and  $-C_{10}$  (coefficient 0.01, enabled in config).

**Gravity vector  $\mathbf{G}(\mathbf{q})$ :**

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix}, \quad G_2 = -(m_1 l_{c1} + m_2 l_1) g \sin \theta_1, \quad G_3 = -m_2 l_{c2} g \sin \theta_2 \quad (5)$$

$G_1 = 0$  because gravity acts perpendicular to the cart's horizontal motion.

**Friction model (viscous + Coulomb):**

$$F_{\text{fric},i} = -b_i \dot{q}_i - \mu_i \text{sign}(\dot{q}_i) \mathbb{1}_{|\dot{q}_i| > 10^{-6}} \quad (6)$$

for each DOF  $i \in \{x, \theta_1, \theta_2\}$ , where  $b_i$  is the viscous coefficient (cart: 0.2, joints: 0.005, 0.004) and  $\mu_i$  is the Coulomb coefficient (from config; varies  $\pm 10\%$  in Monte Carlo). The dead zone  $|\dot{q}_i| > 10^{-6}$  avoids sign-function discontinuity at rest. A viscous-only equation would omit the Coulomb term – the full form (6) is used in all simulations.

*Note on the simplified model for PSO:* The simplified dynamics use a reduced coupling matrix (off-diagonal terms scaled by 0.7–0.8) and omit gyroscopic effects for faster evaluation. The linearization ( $\mathbf{A}, \mathbf{B}$ ) around the upright equilibrium  $\mathbf{q}^* = \mathbf{0}$ ,  $\dot{\mathbf{q}}^* = \mathbf{0}$  is computed numerically via finite differences ( $\varepsilon = 10^{-8}$ ) rather than an analytical closed form, since the symbolic Jacobian of (2) with all coupling terms is unwieldy. PSO uses this linearized model for speed; all validation uses the full nonlinear model.

**1.3 Physical Parameters**

| Parameter           | Symbol     | Value                    |
|---------------------|------------|--------------------------|
| Cart mass           | $M$        | 1.5 kg                   |
| Link 1 mass         | $m_1$      | 0.2 kg                   |
| Link 2 mass         | $m_2$      | 0.15 kg                  |
| Link 1 length       | $l_1$      | 0.4 m                    |
| Link 2 length       | $l_2$      | 0.3 m                    |
| Link 1 COM distance | $l_{c1}$   | 0.2 m                    |
| Link 2 COM distance | $l_{c2}$   | 0.15 m                   |
| Link 1 inertia      | $I_1$      | 0.0081 kg·m <sup>2</sup> |
| Link 2 inertia      | $I_2$      | 0.0034 kg·m <sup>2</sup> |
| Gravity             | $g$        | 9.81 m/s <sup>2</sup>    |
| Cart friction       | $b_c$      | 0.2 N·s/m                |
| Joint 1 friction    | $b_1$      | 0.005 N·m·s/rad          |
| Joint 2 friction    | $b_2$      | 0.004 N·m·s/rad          |
| Max control force   | $u_{\max}$ | 150 N                    |
| Control period      | $\Delta t$ | 0.01 s (100 Hz)          |

**1.4 Model Selection: Simplified vs. Full Nonlinear**

| Model                                  | Use in this project                | Rationale                                                                                                                   |
|----------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Simplified (linearized around upright) | PSO optimization (all controllers) | 10–50× faster per simulation step. PSO requires thousands of evaluations; full model would be prohibitive.                  |
| Full nonlinear (2)                     | All published benchmark results    | Complete Lagrangian, all coupling terms, Coriolis, centrifugal, gyroscopic, full friction. Used for MT-5, MT-7, MT-8, LT-6. |
| Low-rank (SVD approximation)           | HIL mode (hardware-in-the-loop)    | Reduced singular-value truncation preserves stability with lower real-time compute load.                                    |

**Decision rationale:** Controller gain tuning (PSO) uses the simplified model because PSO with 40 particles  $\times$  200 iterations  $\times$  10s simulation on the full nonlinear model would require  $>8$  hours per controller. With the simplified model, PSO completes in minutes. All final validation and reported results use the full nonlinear model.

## 2 Sliding Mode Control: Surface Design and Control Laws

### 2.1 Sliding Surface

The multi-variable sliding surface for the DIP is:

$$\sigma = \lambda_1 \theta_1 + \lambda_2 \theta_2 + k_1 \dot{\theta}_1 + k_2 \dot{\theta}_2 \quad (7)$$

Equivalently, in error coordinates  $e_i = \theta_i - \theta_i^*$  (where  $\theta^* = 0$  at upright equilibrium):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1 \dot{e}_1 + k_2 \dot{e}_2 \quad (8)$$

The surface (7) is a hyperplane in  $\mathbb{R}^4$  (angle + rate space). On this surface ( $\sigma = 0$ ), the reduced dynamics decouple into two independent first-order systems  $\dot{e}_i = -(\lambda_i/k_i) e_i$ , which are stable if and only if:

$$\lambda_i > 0, \quad k_i > 0 \quad (i = 1, 2) \quad (9)$$

These inequalities are enforced as PSO lower bounds (all gains  $> 2$ ).

**Surface design rationale:** The linear form (7) was chosen over alternatives as follows: (i) *Integral sliding surface* – rejected because it adds an integrator state, complicating the Lyapunov analysis for a 6-state system; (ii) *Terminal sliding surface* ( $\sigma = \dot{e} + \beta e^{p/q}$ ) – rejected because the fractional power  $e^{p/q}$  causes numerical issues near the equilibrium ( $p/q < 1$  makes the derivative singular at  $e = 0$ ); (iii) *Linear surface* (7) – relative degree 1 (verified below), Lyapunov analysis straightforward, PSO can tune all 4 surface parameters simultaneously.

**Relative degree check:** Differentiating (7):

$$\dot{\sigma} = \lambda_1 \dot{\theta}_1 + \lambda_2 \dot{\theta}_2 + k_1 \ddot{\theta}_1 + k_2 \ddot{\theta}_2 \quad (10)$$

Substituting  $\ddot{\theta}_i$  from (2):  $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{B}u - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_{\text{fric}})$ . The control  $u$  appears through  $\mathbf{L}\mathbf{M}^{-1}\mathbf{B}$  where  $\mathbf{L} = [0, k_1, k_2]^\top$  (surface gradient w.r.t.  $\ddot{\mathbf{q}}$ ). Since  $\mathbf{M}^{-1}$  is invertible and  $\mathbf{L}\mathbf{M}^{-1}\mathbf{B} \neq 0$  for all physically valid configurations, the relative degree is exactly 1.

**Matching condition:** External disturbances  $d(t)$  and model uncertainty  $\Delta f(\mathbf{x})$  enter the equations of motion through the same actuated direction as  $u$  (the cart force channel):  $\dot{\sigma}$  contains a term  $\mathbf{L}\mathbf{M}^{-1}\mathbf{B}(u + d)$ , so the matching condition holds and the switching gain  $K$  can reject any disturbance bounded by  $|d(t)| \leq \bar{d}$  provided  $K > \bar{d}$ .

### 2.2 Controller Variants: Control Laws

#### 2.2.1 Classical SMC: Saturation Function and Equivalent Control

The saturation function replacing  $\text{sign}(\cdot)$  is:

$$\text{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \leq \varepsilon \\ \text{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \quad (11)$$

Inside the boundary layer ( $|\sigma| \leq \varepsilon$ ), the control is linear in  $\sigma$  – this eliminates high-frequency switching at the cost of a residual steady-state error bounded by  $O(\varepsilon)$ .

The full control law is:

$$u = \underbrace{u_{\text{eq}}}_{\text{equiv. control}} + \underbrace{K \text{ sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d \dot{\sigma}}_{\text{derivative}} \quad (12)$$

**Equivalent control derivation** (from the ideal sliding condition  $\dot{\sigma} = 0$ ):

$$\begin{aligned} \dot{\sigma} = 0 &\implies \mathbf{L}\ddot{\mathbf{q}} + (\lambda_1 \dot{\theta}_1 + \lambda_2 \dot{\theta}_2) = 0 \\ &\implies \mathbf{L}\mathbf{M}^{-1}(\mathbf{B}u_{\text{eq}} - \mathbf{F}) = 0 \\ &\implies u_{\text{eq}} = -\frac{\mathbf{L}\mathbf{M}^{-1}\mathbf{F}}{\mathbf{L}\mathbf{M}^{-1}\mathbf{B}} \end{aligned} \quad (13)$$

where  $\mathbf{F} = \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}}$  collects all nonlinear forcing terms, and  $\mathbf{L} = [0, k_1, k_2]^\top$  is the surface gradient. In code: `LM_inv_B = L @ M_inv @ B` (scalar), `u_eq = -(L @ M_inv @ F) / LM_inv_B`.

- $K$ : switching gain;  $\varepsilon = 0.3$ : boundary layer width (MT-6 study found  $\varepsilon = 0.02$  optimal for chattering only;  $\varepsilon = 0.3$  is the operational value balancing tracking accuracy and compute smoothness)
- 6 gains tuned by PSO:  $[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$

### 2.2.2 Super-Twisting Algorithm (STA-SMC)

The STA operates on the surface derivative, not the surface itself:

$$\begin{aligned} u &= u_1 + u_2 \\ u_1 &= -K_1 |\sigma|^\alpha \text{sign}(\sigma) \quad (\text{continuous}) \\ u_2 &= -K_2 \int_0^t \text{sign}(\sigma(\tau)) d\tau \quad (\text{integral}) \end{aligned} \quad (14)$$

with  $K_1 > K_2 > 0$  and standard exponent  $\alpha = 0.5$ .

**Key property:**  $u_1$  is continuous (chattering reduction), while  $u_2$  handles steady-state errors via integration. Near  $\sigma = 0$ , regularization replaces  $|\sigma|^\alpha$  with  $\delta^\alpha$  (where  $\delta = 10^{-10}$ ) to avoid numerical singularity.

6 gains:  $[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$ ; constraint  $K_1 > K_2$  enforced during PSO.

### 2.2.3 Adaptive SMC

The switching gain adapts online:

$$u = -K(t) \text{ sat}\left(\frac{\sigma}{\varepsilon}\right) \quad (15)$$

$$\dot{K}(t) = \gamma |\sigma| - \delta_{\text{leak}} K(t) \quad \text{with dead zone } |\sigma| < d_z \quad (16)$$

where  $\gamma$  is the adaptation rate,  $\delta_{\text{leak}} = 0.01$  prevents gain windup, and  $d_z = 0.05$  suppresses noise-driven adaptation.  $K(t)$  is clamped to  $[K_{\min}, K_{\max}] = [0.1, 100]$ .

*Parameter rationale:*  $\delta_{\text{leak}} = 0.01$  allows  $K(t)$  to decay slowly when  $\sigma \approx 0$ , preventing accumulation over long horizons. If  $\delta_{\text{leak}}$  is too large,  $K(t)$  shrinks before new disturbances arrive, causing chattering bursts; if too small,  $K(t)$  drifts upward unboundedly. The dead zone  $d_z = 0.05$  matches the angle sensor noise floor ( $0.005 \text{ rad} \times \text{typical gain} \approx 0.01\text{--}0.05$ ), ensuring adaptation does not respond to measurement noise rather than real errors.

5 gains:  $[k_1, k_2, \lambda_1, \lambda_2, \gamma]$ .

### 2.2.4 Hybrid Adaptive STA-SMC

Combines adaptive gain scheduling with the super-twisting structure:

$$u = -k_1(t) |\sigma|^{0.5} \text{sign}(\sigma) - k_2(t) \int \text{sign}(\sigma) dt \quad (17)$$

Adaptation laws for both gains  $k_1(t)$ ,  $k_2(t)$  use rates  $\gamma_1 = 0.1$ ,  $\gamma_2 = 0.05$  with a rate limiter of 5.0/s.

**Known instability (documented):** Under large disturbances and adaptive gain scheduling, this controller produces 666.9° overshoot (176% chattering increase). Root cause: feedback loop between  $k_1(t)$  adaptation and the  $|\sigma|^{0.5}$  term amplifies errors when  $|\sigma| \gg 1$ . This is an open research item.

## 3 PSO Algorithm: Configuration and Methodology

### 3.1 Algorithm Parameters

| Parameter              | Symbol | Value | Rationale                                           |
|------------------------|--------|-------|-----------------------------------------------------|
| Number of particles    | $N_p$  | 40    | Increased from 30 for 6D space                      |
| Iterations             | $T$    | 200   | Convergence verified experimentally                 |
| Inertia weight         | $w$    | 0.7   | Range [0.4, 0.9]; balances exploration/exploitation |
| Cognitive acceleration | $c_1$  | 2.0   | Standard PSO literature value                       |
| Social acceleration    | $c_2$  | 2.0   | Standard PSO literature value                       |
| Random seed            | –      | 42    | Reproducibility                                     |

Velocity update:  $v_{i,d}^{t+1} = w v_{i,d}^t + c_1 r_1 (p_{i,d} - x_{i,d}^t) + c_2 r_2 (g_d - x_{i,d}^t)$ , where  $r_1, r_2 \sim U(0, 1)$ .

### 3.2 Fitness Function

$$J(\boldsymbol{\theta}) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \quad (18)$$

where  $w_1 = 0.7$ ,  $w_2 = 0.3$  (mean vs. worst-case balance), and:

$$J_k = \underbrace{w_{\text{err}} \text{ISE}(\mathbf{x})}_{\text{tracking}} + \underbrace{w_u \text{IAE}(u)}_{\text{effort}} + \underbrace{w_r \text{IAE}(\dot{u})}_{\text{slew rate}} + \underbrace{w_s \int \sigma^2 dt}_{\text{sliding}} + \underbrace{P \cdot \mathbb{1}_{\text{unstable}}}_{\text{penalty}} \quad (19)$$

with weights  $[w_{\text{err}}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$ , normalization thresholds  $[10.0, 100.0, 1000.0, 1.0]$ , and instability penalty  $P = 10^6$ .

### 3.3 Search Bounds per Controller

| Controller           | Surface gains |      | Switch gain |      | Other |     |
|----------------------|---------------|------|-------------|------|-------|-----|
|                      | min           | max  | min         | max  | min   | max |
| Classical SMC (6D)   | 2.0           | 30.0 | 2.0         | 50.0 | 0.05  | 3.0 |
| STA-SMC (6D)         | 2.0           | 30.0 | 2.0         | 30.0 | 0.05  | 3.0 |
| Adaptive SMC (5D)    | 2.0           | 30.0 | 2.0         | 10.0 | 0.05  | 3.0 |
| Hybrid Adaptive (4D) | 2.0           | 30.0 | 2.0         | 10.0 | –     | –   |

Note: STA-SMC enforces  $K_1 > K_2$  as a hard constraint during PSO evaluation (particles violating it are given the instability penalty  $P = 10^6$ ).

*Hyperparameter sensitivity (finding #16, #17):* A formal ablation study over  $N_p$  and  $w$  was not performed; standard literature values were adopted (Kennedy & Eberhart 1995; Clerc & Kennedy 2002).  $N_p = 40$  was increased from 30 to improve coverage of the 6D search space based on the heuristic  $N_p \approx 10d$  for dimension  $d$ . PSO convergence curves are logged to `academic/logs/ps0/` and show stable cost plateau by iteration 150–180 in all controllers; the 200-iteration budget is fixed (no early-stopping criterion implemented).

*Cost function scale calibration (finding #18):* Normalization thresholds  $[10, 100, 1000, 1]$  correspond to the approximate magnitude of each raw metric for an unstable run:  $\text{ISE} \sim 10 \text{ rad}^2$ ,  $\text{IAE}(\mathbf{u}) \sim 100 \text{ N}\cdot\text{s}$ ,  $\text{IAE}(\dot{\mathbf{u}}) \sim 1000 \text{ N}$ , sliding integral  $\sim 1$ . The instability penalty  $P = 10^6$  is approximately  $10^4\times$  the typical stable-run cost ( $\sim 0.1\text{--}1$ ), ensuring unstable particles are always ranked last in the swarm.

### 3.4 Robust PSO: Scenario-Based Optimization

Standard PSO (nominal initial condition) produces gains that fail catastrophically on varied initial conditions:

$$\text{Nominal chattering: } 2.14 \pm 0.13 \quad \longrightarrow \quad \text{Realistic scenarios: } 107.61 \pm 5.48 \quad (\times 50.4 \text{ degradation}) \quad (20)$$

**Robust PSO solution:** Evaluate each particle on  $N_s = 15$  scenarios:

- 3 nominal:  $\theta_0 \in [-0.05, +0.05]$  rad
- 4 moderate:  $\theta_0 \in [-0.15, +0.15]$  rad
- 8 large:  $\theta_0 \in [-0.30, +0.30]$  rad

Fitness:  $J_{\text{robust}} = 0.7 \bar{J} + 0.3 \max J_k$  (weighted average + worst-case, same formula as (18)).

**Result:** Chattering degradation reduced from  $50.4\times$  to  $7.5\times$  ( $144.59\times \rightarrow 19.28\times$  in absolute terms).

*Scenario distribution rationale (finding #19):* The 3+4+8 split (20%, 27%, 53% nominal/moderate/large) was chosen to weight large perturbations more heavily than nominal, since controllers failing at large initial angles are useless in practice. A sensitivity analysis on this split was not performed; this is a known limitation.

### 3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

| Controller |           | $k_1$ | $k_2$ | $\lambda_1$ | $\lambda_2$       | Gain 5        | Gain 6           | Cost     |
|------------|-----------|-------|-------|-------------|-------------------|---------------|------------------|----------|
| Classical  | SMC       | 5.0   | 5.0   | 5.0         | 0.5               | 0.5           | 0.5              | baseline |
| (nom.)     |           |       |       |             |                   |               |                  |          |
| Classical  | SMC (ro-  | 23.07 | 12.85 | 5.51        | 3.49              | 2.23          | 0.15             | -2.2%    |
| bust)      |           |       |       |             |                   |               |                  |          |
| STA-SMC    | (nominal) | 12.0  | 6.0   | 4.85        | 3.43              | $K_1=8.0$     | $K_2=4.0$        | baseline |
| STA-SMC    | (robust)  | 5.62  | 3.75  | 4.36        | 2.05              | $K_1=2.02$    | $K_2=6.67$       | –        |
| Adaptive   | SMC       | 10.0  | 8.0   | 5.0         | 4.0               | $\gamma=1.0$  | –                | baseline |
| Adaptive   | SMC (ro-  | 2.14  | 3.36  | 7.20        | 0.34              | $\gamma=0.29$ | –                | –        |
| bust)      |           |       |       |             |                   |               |                  |          |
| Hybrid     | (robust)  | –     | –     | $c_1=10.15$ | $\lambda_1=12.84$ | $c_2=6.82$    | $\lambda_2=2.75$ | -21.4%   |

**Known inconsistency – STA robust gains (finding #20):** The robust STA gains show  $K_1 = 2.02$ ,  $K_2 = 6.67$ , which violates the stated constraint  $K_1 > K_2$ . This occurred because the robust PSO cost function (15 scenarios, worst-case weighted) found this parameter combination



reduced chattering across the scenario set even though the nominal stability condition is reversed. The Moreno–Osorio proof requires  $K_1 > K_2 > 0$  for finite-time convergence; these gains are therefore *not theoretically guaranteed* and should be used with caution. This is an open issue requiring either re-running robust PSO with the hard constraint enforced, or re-deriving stability conditions for  $K_1 < K_2$ .

## 4 Lyapunov Stability: Proofs and Validation

Proofs are documented in `academic/paper/sphinx_docs/theory/lyapunov_stability_proofs.md` (v2.0, 1427 lines, 9 theorems + 5 lemmas, 15 BibTeX references).

*Note on symbolic constants (finding #21):* The constants  $\beta$ ,  $\eta$ ,  $c_1$ ,  $c_2$ ,  $\alpha_1$ ,  $\alpha_2$  in the inequalities below are **not fixed numerical values** – they are state-dependent or design-parameter-dependent quantities. Specifically:  $\beta = \mathbf{LM}^{-1}(\mathbf{q})\mathbf{B} > 0$  (varies with configuration  $\mathbf{q}$ ; positive for all valid DIP poses);  $\eta = K - \bar{d}$  (switching gain minus disturbance bound; positive by design if  $K > \bar{d}$ ). The proofs establish that these quantities remain positive along all trajectories, which is the content of the theorems – not that they have fixed values.

### 4.1 Classical SMC

$$V = \frac{1}{2}\sigma^2 \quad (21)$$

$$\dot{V} = \sigma\dot{\sigma} \leq -\beta\eta|\sigma| - \beta k_d\sigma^2 < 0 \quad (22)$$

for all  $|\sigma| > \varepsilon$  (outside boundary layer), where  $\beta = \mathbf{LM}^{-1}\mathbf{B} > 0$  and  $\eta = K - \bar{d} > 0$ . Reaching time bound:  $t_{\text{reach}} \leq |\sigma(0)|/(\eta\beta)$ . Stability type: **exponential asymptotic** (when  $k_d > 0$ ).

**Numerical validation:** 96.2% of simulation samples satisfy  $\dot{V} < 0$  outside the boundary layer. The remaining 3.8% occur during the initial reaching phase when the trajectory is still crossing the boundary layer ( $|\sigma| \leq \varepsilon = 0.3$ ) – this is expected behavior and does not indicate instability.

### 4.2 Super-Twisting (STA-SMC)

Uses the generalized gradient Lyapunov function (Moreno & Osorio 2008):

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \quad z = u_2 \quad (23)$$

$$\dot{V}_{\text{STA}} \leq -c_1\|\boldsymbol{\xi}\|^{3/2} + c_2L \quad (24)$$

where  $\boldsymbol{\xi} = [|\sigma|^{1/2}\text{sign}(\sigma), z]^\top$ ,  $L$  is the Lipschitz constant of  $\dot{d}_u(t)$  (disturbance derivative), and  $c_1, c_2 > 0$  depend on  $K_1, K_2$ .

**Moreno–Osorio gain conditions** (required for finite-time convergence):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \quad K_2 > \frac{\bar{d}}{\beta}, \quad K_1 > K_2 \quad (25)$$

where  $\bar{d}$  bounds the disturbance and  $\beta = \mathbf{LM}^{-1}\mathbf{B}$ . These conditions are satisfied by the nominal gains ( $K_1 = 8 > K_2 = 4$ ) but *not* by the robust gains ( $K_1 = 2.02 < K_2 = 6.67$ ) – see the note in Section 3.5.

**Finite-time convergence:** The STA guarantees  $\sigma(t) = 0$  for all  $t \geq T_f$ , where  $T_f$  depends on initial conditions and gain values. An explicit closed-form bound for the DIP system is not derived here (it requires bounding  $\beta$  over the state trajectory, which is configuration-dependent); the experimental confirmation is convergence in 1.82 s for typical initial conditions.

Stability type: **finite-time convergence**.

**Numerical validation:** Convergence in 1.82 s (16% faster than Classical SMC at 2.15 s).

### 4.3 Adaptive SMC

Composite Lyapunov (state + parameter estimation error):

$$V_{\text{ad}} = \frac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \quad \tilde{K} = K(t) - K^* \quad (26)$$

$$\dot{V}_{\text{ad}} \leq -\eta|\sigma| \Rightarrow \sigma(t) \rightarrow 0, \quad K(t) \text{ bounded} \quad (27)$$

Stability type: **asymptotic with bounded adaptive gain**.

**Numerical validation:** 100% of simulation samples within  $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$ .

### 4.4 Hybrid Adaptive STA-SMC

Extended composite Lyapunov:

$$V_{\text{hyb}} = \frac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \frac{1}{2}u_{\text{int}}^2 \quad (28)$$

$$\dot{V}_{\text{hyb}} \leq -\alpha_1 V_{\text{hyb}} + \alpha_2 \|\mathbf{w}\| \quad (29)$$

Stability type: **Input-to-State Stable (ISS)**.

**Numerical validation:** All signals bounded, 0 emergency resets during nominal conditions.

### 4.5 Swing-Up SMC

Switched Lyapunov approach for the two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & (\text{swing-up phase}) \\ \frac{1}{2}\sigma^2 & (\text{stabilization phase}) \end{cases} \quad (30)$$

Global stability proven; hysteresis in switching prevents Zeno behavior (finite switching guaranteed).

### 4.6 MPC

Lyapunov function is the optimal cost-to-go  $V_k = J_k^*$ :

$$V_{k+1} - V_k \leq -\lambda_{\min}(\mathbf{Q})\|\mathbf{x}_k\|^2 < 0 \quad (31)$$

Stability type: **asymptotic** (requires recursive feasibility).

#### 4.7 Summary Table

| Controller                                        | Lyapunov $V$                                | Stability                                             | Settling (s) |
|---------------------------------------------------|---------------------------------------------|-------------------------------------------------------|--------------|
| Classical SMC                                     | $\frac{1}{2}\sigma^2$                       | Asymptotic (exp.)                                     | 2.15         |
| STA-SMC                                           | $ \sigma  + z^2/2K_2$                       | Finite-time                                           | 1.82         |
| Adaptive SMC                                      | $\sigma^2 + \tilde{K}^2/\gamma$             | Asymptotic (bnd.)                                     | 2.35         |
| Hybrid Adaptive STA                               | $\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$ | ISS                                                   | 1.95         |
| Swing-Up                                          | Energy-based (switched)                     | Global                                                | –            |
| MPC                                               | Optimal cost $J^*$                          | Asymptotic                                            | –            |
| <i>Convergence ordering (fastest to slowest):</i> |                                             | STA $\dot{}$ Hyb $\dot{}$ Classical $\dot{}$ Adaptive |              |

## 5 Monte Carlo Benchmark Results

### 5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs per controller (MT-5 study)
- **Model:** Full nonlinear dynamics (2)
- **Initial condition distribution** (uniform, independent):
  - Cart position  $x_0 \sim U(-0.05, +0.05)$  m
  - Angles  $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$  rad ( $\approx \pm 2.9^\circ$ )
  - Cart velocity  $\dot{x}_0 \sim U(-0.02, +0.02)$  m/s
  - Angular velocities  $\dot{\theta}_{1,0}, \dot{\theta}_{2,0} \sim U(-0.05, +0.05)$  rad/s
- **Success criterion:**  $|\theta_i(t)| \leq 2\%$  of equilibrium maintained for at least 1 continuous second within the 10 s window. All 4 tested controllers achieved 100% success (no divergence).
- **Controller inclusion:** 4 of 7 controllers tested. Excluded: Swing-Up (diverges in sliding phase with these ICs), MPC (requires cvxpy; not integrated in batch runner), Factory wrapper (not a controller).
- **Simulation duration:** 10 s at  $\Delta t = 0.01$  s (1000 steps)
- **Total simulations:** 400 (4 controllers  $\times$  100 runs)
- **Random seed:** Not documented in MT-5 reports – this is a reproducibility gap; future runs should log seed.
- **Statistics:** Bootstrap 95% CI (10,000 resamples); Welch’s  $t$ -test (does not require normality assumption for  $n = 100$ ); no Bonferroni correction applied (all comparisons were pre-planned). Cohen’s  $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$  with pooled SD.

## 5.2 Primary Performance Metrics

| Controller                | Settling (s)    | Overshoot (%) | Energy (N <sup>2</sup> s) | Chattering        |
|---------------------------|-----------------|---------------|---------------------------|-------------------|
| Classical SMC             | $2.15 \pm 0.18$ | $5.8 \pm 0.8$ | $9,843 \pm 7,518$         | $0.647 \pm 0.347$ |
| STA-SMC                   | $1.82 \pm 0.15$ | $2.3 \pm 0.4$ | $202,907 \pm 15,749$      | $3.088 \pm 0.141$ |
| Adaptive SMC              | $2.35 \pm 0.21$ | $8.2 \pm 1.1$ | $214,255 \pm 6,254$       | $3.098 \pm 0.030$ |
| Hybrid Adap. <sup>†</sup> | –               | –             | $10^6$ (sentinel)         | 0.0 (sentinel)    |

Values: mean  $\pm$  SD; 95% CI in Table below.

<sup>†</sup> Hybrid returned sentinel values (energy= $10^6$ , chattering=0) on all 100 runs.

This indicates internal controller failure (gain divergence), not simulation abort.

| Controller    | Settling 95% CI | Overshoot 95% CI | Compute ( $\mu$ s) | Score |
|---------------|-----------------|------------------|--------------------|-------|
| Classical SMC | [1.97, 2.33] s  | [5.0, 6.6]%      | $18.5 \pm 2.1$     | 85.1% |
| STA-SMC       | [1.67, 1.97] s  | [1.9, 2.7]%      | $24.2 \pm 3.5$     | 92.5% |
| Adaptive SMC  | [2.14, 2.56] s  | [7.1, 9.3]%      | $31.6 \pm 4.2$     | 78.4% |
| Hybrid Adap.  | [1.79, 2.11] s  | [3.0, 4.0]%      | $26.8 \pm 3.1$     | 88.3% |

All controllers meet the 100  $\mu$ s real-time deadline (68–81% headroom).

## 5.3 Chattering Analysis (FFT)

| Controller    | Peak freq. (Hz) | Chattering index | Energy in chatt. band | vs. Classical |
|---------------|-----------------|------------------|-----------------------|---------------|
| Classical SMC | $\approx 35$    | 8.2              | 12.3%                 | baseline      |
| STA-SMC       | $\approx 8$     | 2.1              | 2.1%                  | –74%          |
| Hybrid Adap.  | $\approx 28$    | 5.4              | 8.5%                  | –34%          |
| Adaptive SMC  | $\approx 42$    | 9.7              | 15.1%                 | +18%          |

**STA chattering mechanism:** Moving the switching action to the integral  $u_2$  (inside the control loop) vs. directly to  $u$  in classical SMC reduces the high-frequency switching amplitude from 35 Hz to 8 Hz in the output.

**Metric disambiguation (finding #31):** Two different chattering metrics appear in this report and should not be compared directly:

- *Chattering amplitude* (MT-5, primary metrics table): RMS of the control signal  $u(t)$  over the full 10 s window. Classical: 0.647; STA: 3.088. STA is *higher* here because its integral term  $u_2$  accumulates larger absolute values.
- *Chattering index* (QW-2, FFT table above): RMS amplitude in the high-frequency band (>10 Hz) of  $u(t)$ . Classical: 8.2; STA: 2.1. STA is *lower* because the high-frequency content is reduced by the continuous  $u_1$  term.

These metrics measure different phenomena. For actuator wear, the FFT chattering index is more physically meaningful.

#### 5.4 Statistical Significance (Welch’s $t$ -test)

| Comparison            | $t$ -stat | $p$ -value | Cohen’s $d$ | Result                          |
|-----------------------|-----------|------------|-------------|---------------------------------|
| <i>Settling time:</i> |           |            |             |                                 |
| STA vs. Classical     | 2.14      | 0.038      | 2.14        | Significant (−16%, −330 ms)     |
| STA vs. Adaptive      | 3.67      | 0.002      | –           | Highly significant (−29%)       |
| STA vs. Hybrid        | 1.03      | 0.312      | –           | Not significant (practical tie) |
| Hybrid vs. Adaptive   | 2.31      | 0.031      | –           | Significant (−17%)              |
| <i>Overshoot:</i>     |           |            |             |                                 |
| STA vs. Classical     | 4.89      | <0.001     | –           | Highly significant (−60%)       |
| STA vs. Adaptive      | 6.42      | <0.001     | –           | Highly significant (−72%)       |
| STA vs. Hybrid        | 2.34      | 0.030      | –           | Significant (−34%)              |

Cohen’s  $d = 2.14$  for STA vs. Classical (settling time) indicates a *large* practical effect size, not just statistical significance.

#### 5.5 Energy Trade-off Analysis

| Energy phase (Classical SMC) | Energy        | Share       |
|------------------------------|---------------|-------------|
| Reaching phase (0–0.5 s)     | 6.2 J         | 50%         |
| Sliding phase (0.5–2.1 s)    | 5.8 J         | 47%         |
| Steady state (> 2.1 s)       | 0.4 J         | 3%          |
| <b>Classical total</b>       | <b>12.4 J</b> | –           |
| <b>STA total</b>             | 11.8 J        | – 4.8% less |
| <b>Adaptive total</b>        | 13.6 J        | – 9.7% more |

**Note on MT-5 energy values:** The MT-5 study reports energy in  $\text{N}^2\cdot\text{s}$  (integral of  $u^2$ , control effort metric). The QW-2 single-run study reports in Joules (physical energy). Both are valid metrics; the MT-5  $\text{N}^2\cdot\text{s}$  values show a  $20\times$  difference between Classical and STA because the MT-5 initial conditions span a wider range (higher control effort required).

## 6 Boundary Layer Study: MT-6 Honest Result

The boundary layer width  $\varepsilon$  in (12) controls the chattering/accuracy tradeoff.

| $\varepsilon$ | Chattering ( $\Delta$ from baseline) | Overshoot | Settling | Notes                   |
|---------------|--------------------------------------|-----------|----------|-------------------------|
| 0.01          | baseline (high)                      | nominal   | nominal  | Too narrow, noisy       |
| 0.02          | −3.7% (optimal)                      | nominal   | nominal  | Near-optimal            |
| 0.05          | −10%                                 | +7.5%     | degraded | Too wide, accuracy loss |

**Important correction:** An initial analysis reported 66.5% chattering reduction from adaptive boundary layer  $\varepsilon(t) = \varepsilon_{\min} + \kappa|\sigma|$ . A frequency-domain re-analysis showed this was a *transient metric artifact* (the initial transient dominated the RMS computation). The unbiased frequency-domain metric gives only 3.7%.

**Conclusion:** Fixed  $\varepsilon = 0.02$  is near-optimal for the DIP. Adaptive boundary layers are not justified by this result.

*Study limitations (finding #28):* Only 3 values of  $\varepsilon$  were tested – a finer sweep was not performed. The study was run on the **full nonlinear model** (2). The corrected metric uses

FFT-based RMS in the high-frequency band ( $>10$  Hz) after removing the initial 0.5 s transient; the biased original metric included the transient in the RMS window, inflating chattering reduction to 66.5%.

## 7 Robustness Analysis

### 7.1 Model Uncertainty (LT-6)

All controllers tested under parametric uncertainty:  $\pm 10\%$  and  $\pm 20\%$  on all physical parameters (cart mass, link masses, lengths, inertias). Friction coefficients varied by  $\pm 10\%$ .

*Measurement protocol (finding #29):* All parameters were perturbed **simultaneously** (worst-case combination, not one at a time). “Mismatch tolerance” is defined as the largest simultaneous uniform perturbation factor  $\delta$  such that the controller remains stable (does not diverge within 10 s) under the perturbed model. For example, Classical SMC at  $\pm 12\%$  means: stable for  $\delta \leq 12\%$ , but first instability observed between 12% and the next tested level (15%). A finer sweep to find the exact stability boundary was not performed.

| Controller          | $\pm 10\%$ mismatch | $\pm 20\%$ mismatch |                        |
|---------------------|---------------------|---------------------|------------------------|
| Classical SMC       | Stable              | Stable              | 12% mismatch tolerance |
| STA-SMC             | Stable              | Stable              | 8% mismatch tolerance  |
| Adaptive SMC        | Stable              | Stable              | 15% mismatch tolerance |
| Hybrid Adaptive STA | Stable              | Stable              | <b>16% (best)</b>      |

The Adaptive and Hybrid controllers tolerate larger uncertainty because the adaptive gain  $K(t)$  compensates for the unknown bounds automatically (consistent with the ISS Lyapunov proof in Section 4.4).

### 7.2 Disturbance Rejection (MT-8)

External step disturbances applied to cart; attenuation measured as  $1 - \|\mathbf{x}_{\text{dist}}\|/\|\mathbf{x}_{\text{nom}}\|$ :

| Controller          | Disturbance attenuation |
|---------------------|-------------------------|
| STA-SMC             | <b>91% (best)</b>       |
| Hybrid Adaptive STA | 89%                     |
| Classical SMC       | 87%                     |
| Adaptive SMC        | 78%                     |

STA-SMC’s continuous  $u_1$  component (14) provides better disturbance rejection than the discontinuous sign function in Classical SMC.

## 8 Thesis and Paper Status

### 8.1 Research Paper (LT-7): Submission-Ready

| Item               | Value                                                                   |
|--------------------|-------------------------------------------------------------------------|
| Status             | <b>[DONE]</b> SUBMISSION-READY (v2.1, Dec 25 2025)                      |
| Pages              | 71 pages                                                                |
| Figures            | 14 (all 300 DPI, auto-generated from raw data)                          |
| Simulation total   | 1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty) |
| Target journal (1) | <i>IEEE Trans. Control Systems Technology</i>                           |
| Target journal (2) | <i>IFAC World Congress / IEEE CDC</i>                                   |

### 8.2 Thesis: 90/100 (Near Submission-Ready)

| Phase                           | Target score                                                | Status             | Target date |
|---------------------------------|-------------------------------------------------------------|--------------------|-------------|
| Phases 1–2 (figures, tables)    | +20 pts                                                     | <b>[DONE]</b> Done | Dec 29 2025 |
| Phase 3 (bibliography +15 refs) | +2 pts                                                      | <b>[PENDING]</b>   | Jan 10 2026 |
| Phase 4 (proofreading)          | +3 pts                                                      | <b>[PENDING]</b>   | Jan 13 2026 |
| Phase 5 (final QA)              | +3 pts                                                      | <b>[PENDING]</b>   | Jan 14 2026 |
| Phase 6 (advisor review)        | +2 pts                                                      | <b>[PENDING]</b>   | Jan 24 2026 |
| <b>Current: 90/100</b>          | <b>Target: 100/100 by Jan 24, 2026 (17 hours remaining)</b> |                    |             |

Completed figures: system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB). All meet 300 DPI and 500 KB requirements.

## A Controller Gains Reference (PSO-Tuned, Operational)

| Controller          | Gain                                         | Nominal                     | Robust (MT-8)                            |
|---------------------|----------------------------------------------|-----------------------------|------------------------------------------|
| Classical SMC       | $[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$   | $[5, 5, 5, 0.5, 0.5, 0.5]$  | $[23.07, 12.85, 5.51, 3.49, 2.23, 0.15]$ |
| STA-SMC             | $[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$ | $[12, 6, 4.85, 3.43, 8, 4]$ | $[5.62, 3.75, 4.36, 2.05, 2.02, 6.67]$   |
| Adaptive SMC        | $[k_1, k_2, \lambda_1, \lambda_2, \gamma]$   | $[10, 8, 5, 4, 1]$          | $[2.14, 3.36, 7.20, 0.34, 0.29]$         |
| Hybrid Adaptive STA | $[c_1, \lambda_1, c_2, \lambda_2]$           | $[5, 5, 5, 0.5]$            | $[10.15, 12.84, 6.82, 2.75]$             |

## B Key Citations

- Utkin, V. I. (1992). *Sliding Modes in Control and Optimization*.
- Slotine, J.-J. E. & Li, W. (1991). *Applied Nonlinear Control*.
- Levant, A. (2003). Higher-order sliding modes. *IEEE TAC*, 48(6).
- Moreno, J. A. & Osorio, M. (2008). A Lyapunov approach to second-order sliding modes. *IFAC Proceedings*.

- Kennedy, J. & Eberhart, R. C. (1995). Particle swarm optimization. *Proc. IEEE ICNN*.
- Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.).